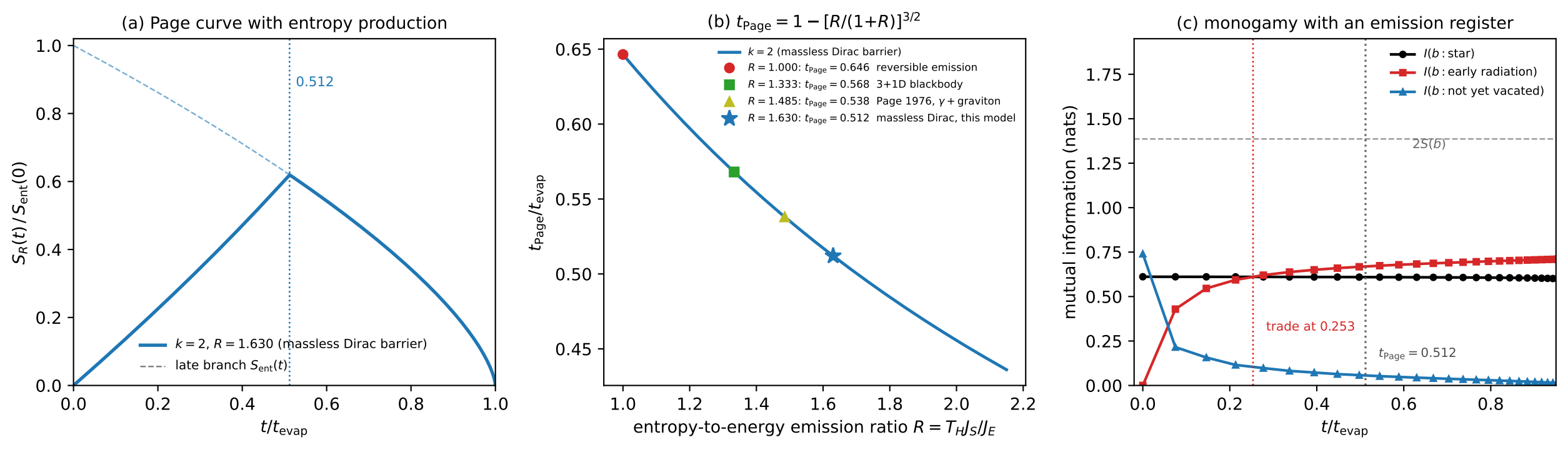
(a) Page curve with entropy production
0.512
$k = 2$, $R = 1.630$ (massless Dirac barrier)
late branch $S_{ent}(t)$
$S_R(t) / S_{ent}(0)$
$t/t_{evap}$
(b) $t_{Page} = 1 - [R/(1+R)]^{3/2}$
$k = 2$ (massless Dirac barrier)
$R = 1.000$: $t_{Page} = 0.646$ reversible emission
$R = 1.333$: $t_{Page} = 0.568$ 3+1D blackbody
$R = 1.485$: $t_{Page} = 0.538$ Page 1976, $\gamma$ + graviton
$R = 1.630$: $t_{Page} = 0.512$ massless Dirac, this model
$t_{Page}/t_{evap}$
entropy-to-energy emission ratio $R = T_H J_S / J_E$
(c) monogamy with an emission register
$I(b : star)$
$I(b : early radiation)$
$I(b : not yet vacated)$
$2S(b)$
trade at 0.253
$t_{Page} = 0.512$
mutual information (nats)
$t/t_{evap}$